

1 Product formula for the Jacobi theta function

In its most elaborate form, Jacobi's **theta function** is defined for $z \in \mathbb{C}$ and $\tau \in \mathbb{H}$ by

$$(1) \quad \Theta(z|\tau) = \sum_{n=-\infty}^{\infty} e^{\pi i n^2 \tau} e^{2\pi i n z}.$$

Two significant special cases (or variants) are $\theta(\tau)$ and $\vartheta(t)$, which are defined by

$$\begin{aligned} \theta(\tau) &= \sum_{n=-\infty}^{\infty} e^{\pi i n^2 \tau}, \quad \tau \in \mathbb{H}, \\ \vartheta(t) &= \sum_{n=-\infty}^{\infty} e^{-\pi n^2 t}, \quad t > 0. \end{aligned}$$

In fact, the relation between these various functions is given by $\theta(\tau) = \Theta(0|\tau)$ and $\vartheta(t) = \theta(it)$, with of course, $t > 0$.

We have already encountered these functions several times. For example, in the study of the heat diffusion equation for the circle, in Chapter 4 of Book I, we found that the heat kernel was given by

$$H_t(x) = \sum_{n=-\infty}^{\infty} e^{-4\pi^2 n^2 t} e^{2\pi i n x},$$

and therefore $H_t(x) = \Theta(x|4\pi i t)$.

Another instance was the occurrence of ϑ in the study of the zeta function. In fact, we proved in Chapter 6 that the functional equation of ϑ implied that of ζ , which then led to the analytic continuation of the zeta function.

We begin our closer look at Θ as a function of z , with τ fixed, by recording its basic structural properties, which to a large extent characterize it.

Proposition 1.1 *The function Θ satisfies the following properties:*

- (i) Θ is entire in $z \in \mathbb{C}$ and holomorphic in $\tau \in \mathbb{H}$.
- (ii) $\Theta(z+1|\tau) = \Theta(z|\tau)$.
- (iii) $\Theta(z+\tau|\tau) = \Theta(z|\tau)e^{-\pi i \tau}e^{-2\pi i z}$.
- (iv) $\Theta(z|\tau) = 0$ whenever $z = 1/2 + \tau/2 + n + m\tau$ and $n, m \in \mathbb{Z}$.

Proof. Suppose that $\text{Im}(\tau) = t \geq t_0 > 0$ and $z = x + iy$ belongs to a bounded set in $\mathbb{C}$, say $|z| \leq M$. Then, the series defining Θ is absolutely and uniformly convergent, since

$$\sum_{n=-\infty}^{\infty} |e^{\pi i n^2 \tau} e^{2\pi i n z}| \leq C \sum_{n \geq 0} e^{-\pi n^2 t_0} e^{2\pi n M} < \infty.$$

Therefore, for each fixed $\tau \in \mathbb{H}$ the function $\Theta(\cdot|\tau)$ is entire, and for each fixed $z \in \mathbb{C}$ the function $\Theta(z|\cdot)$ is holomorphic in the upper half-plane.

Since the exponential $e^{2\pi i n z}$ is periodic of period 1, property (ii) is immediate from the definition of Θ .

To show the third property we may complete the squares in the expression for $\Theta(z + \tau|\tau)$. In detail, we have

$$\begin{aligned} \Theta(z + \tau|\tau) &= \sum_{n=-\infty}^{\infty} e^{\pi i n^2 \tau} e^{2\pi i n(z+\tau)} \\ &= \sum_{n=-\infty}^{\infty} e^{\pi i(n^2+2n)\tau} e^{2\pi i n z} \\ &= \sum_{n=-\infty}^{\infty} e^{\pi i(n+1)^2 \tau} e^{-\pi i \tau} e^{2\pi i n z} \\ &= \sum_{n=-\infty}^{\infty} e^{\pi i(n+1)^2 \tau} e^{-\pi i \tau} e^{2\pi i(n+1)z} e^{-2\pi i z} \\ &= \Theta(z|\tau) e^{-\pi i \tau} e^{-2\pi i z}. \end{aligned}$$

Thus we see that $\Theta(z|\tau)$, as a function of z , is periodic with period 1 and “quasi-periodic” with period τ .

To establish the last property it suffices, by what was just shown, to prove that $\Theta(1/2 + \tau/2|\tau) = 0$. Again, we use the interplay between n and n^2 to get

$$\begin{aligned} \Theta(1/2 + \tau/2|\tau) &= \sum_{n=-\infty}^{\infty} e^{\pi i n^2 \tau} e^{2\pi i n(1/2+\tau/2)} \\ &= \sum_{n=-\infty}^{\infty} (-1)^n e^{\pi i(n^2+n)\tau}. \end{aligned}$$

To see that this last sum is identically zero, it suffices to match $n \geq 0$ with $-n-1$, and to observe that they have opposite parity, and that $(-n-1)^2 + (-n-1) = n^2 + n$. This completes the proof of the proposition.

We consider next a product $\Pi(z|\tau)$ that enjoys the same structural properties as $\Theta(z|\tau)$ as a function of z . This product is defined for $z \in \mathbb{C}$ and $\tau \in \mathbb{H}$ by

$$\Pi(z|\tau) = \prod_{n=1}^{\infty} (1 - q^{2n})(1 + q^{2n-1}e^{2\pi iz})(1 + q^{2n-1}e^{-2\pi iz}),$$

where we have used the notation that is standard in the subject, namely $q = e^{\pi i\tau}$. The function $\Pi(z|\tau)$ is sometimes referred to as the **triple-product**.

Proposition 1.2 *The function $\Pi(z|\tau)$ satisfies the following properties:*

- (i) $\Pi(z, \tau)$ is entire in $z \in \mathbb{C}$ and holomorphic for $\tau \in \mathbb{H}$.
- (ii) $\Pi(z + 1|\tau) = \Pi(z|\tau)$.
- (iii) $\Pi(z + \tau|\tau) = \Pi(z|\tau)e^{-\pi i\tau}e^{-2\pi iz}$.
- (iv) $\Pi(z|\tau) = 0$ whenever $z = 1/2 + \tau/2 + n + m\tau$ and $n, m \in \mathbb{Z}$. Moreover, these points are simple zeros of $\Pi(\cdot|\tau)$, and $\Pi(\cdot|\tau)$ has no other zeros.

Proof. If $\text{Im}(\tau) = t \geq t_0 > 0$ and $z = x + iy$, then $|q| \leq e^{-\pi t_0} < 1$ and

$$(1 - q^{2n})(1 + q^{2n-1}e^{2\pi iz})(1 + q^{2n-1}e^{-2\pi iz}) = 1 + O(|q|^{2n-1}e^{2\pi|z|}).$$

Since the series $\sum |q|^{2n-1}$ converges, the results for infinite products in Chapter 5 guarantee that $\Pi(z|\tau)$ defines an entire function of z with $\tau \in \mathbb{H}$ fixed, and a holomorphic function for $\tau \in \mathbb{H}$ with $z \in \mathbb{C}$ fixed.

Also, it is clear from the definition that $\Pi(z|\tau)$ is periodic of period 1 in the z variable.

To prove the third property, we first observe that since $q^2 = e^{2\pi i\tau}$ we have

$$\begin{aligned} \Pi(z + \tau|\tau) &= \prod_{n=1}^{\infty} (1 - q^{2n})(1 + q^{2n-1}e^{2\pi i(z+\tau)})(1 + q^{2n-1}e^{-2\pi i(z+\tau)}) \\ &= \prod_{n=1}^{\infty} (1 - q^{2n})(1 + q^{2n+1}e^{2\pi iz})(1 + q^{2n-3}e^{-2\pi iz}). \end{aligned}$$

Comparing this last product with $\Pi(z|\tau)$, and isolating the factors that are either missing or extra leads to

$$\Pi(z + \tau|\tau) = \Pi(z|\tau) \left(\frac{1 + q^{-1}e^{-2\pi iz}}{1 + qe^{2\pi iz}} \right).$$

Hence (iii) follows because $(1+x)/(1+x^{-1}) = x$, whenever $x \neq -1$.

Finally, to find the zeros of $\Pi(z|\tau)$ we recall that a product that converges vanishes only if at least one of its factors is zero. Clearly, the factor $(1 - q^n)$ never vanishes since $|q| < 1$. The second factor $(1 + q^{2n-1}e^{2\pi iz})$ vanishes when $q^{2n-1}e^{2\pi iz} = -1 = e^{\pi i}$. Since $q = e^{\pi i\tau}$, we then have¹

$$(2n-1)\tau + 2z = 1 \pmod{2}.$$

Hence,

$$z = 1/2 + \tau/2 - n\tau \pmod{1},$$

and this takes care of the zeros of the type $1/2 + \tau/2 - n\tau + m$ with $n \geq 1$ and $m \in \mathbb{Z}$. Similarly, the third factor vanishes if

$$(2n-1)\tau - 2z = 1 \pmod{2}$$

which implies that

$$\begin{aligned} z &= -1/2 - \tau/2 + n\tau \pmod{1} \\ &= 1/2 + \tau/2 + n'\tau \pmod{1}, \end{aligned}$$

where $n' \geq 0$. This exhausts the zeros of $\Pi(\cdot|\tau)$. Finally, these zeros are simple, since the function $e^w - 1$ vanishes at the origin to order 1 (a fact obvious from a power series expansion or a simple differentiation).

The importance of the product Π comes from the following theorem, called the product formula for the theta function. The fact that $\Theta(z|\tau)$ and $\Pi(z|\tau)$ satisfy similar properties hints at a close connection between the two. This is indeed the case.

Theorem 1.3 (Product formula) *For all $z \in \mathbb{C}$ and $\tau \in \mathbb{H}$ we have the identity $\Theta(z|\tau) = \Pi(z|\tau)$.*

Proof. Fix $\tau \in \mathbb{H}$. We claim first that there exists a constant $c(\tau)$ such that

$$(2) \quad \Theta(z|\tau) = c(\tau)\Pi(z|\tau).$$

In fact, consider the quotient $F(z) = \Theta(z|\tau)/\Pi(z|\tau)$, and note that by the previous two propositions, the function F is entire and doubly periodic with periods 1 and τ . This implies that F is constant as claimed.

¹We use the standard short-hand, $a = b \pmod{c}$, to mean that $a - b$ is an integral multiple of c .

We must now prove that $c(\tau) = 1$ for all τ , and the main point is to establish that $c(\tau) = c(4\tau)$. If we put $z = 1/2$ in (2), so that $e^{2i\pi z} = e^{-2i\pi z} = -1$, we obtain

$$\begin{aligned} \sum_{n=-\infty}^{\infty} (-1)^n q^{n^2} &= c(\tau) \prod_{n=1}^{\infty} (1 - q^{2n})(1 - q^{2n-1})(1 - q^{2n-1}) \\ &= c(\tau) \prod_{n=1}^{\infty} [(1 - q^{2n-1})(1 - q^{2n})] (1 - q^{2n-1}) \\ &= c(\tau) \prod_{n=1}^{\infty} (1 - q^n)(1 - q^{2n-1}). \end{aligned}$$

Hence

$$(3) \quad c(\tau) = \frac{\sum_{n=-\infty}^{\infty} (-1)^n q^{n^2}}{\prod_{n=1}^{\infty} (1 - q^n)(1 - q^{2n-1})}.$$

Next, we put $z = 1/4$ in (2), so that $e^{2i\pi z} = i$. On the one hand, we have

$$\Theta(1/4|\tau) = \sum_{n=-\infty}^{\infty} q^{n^2} i^n,$$

and due to the fact that $1/i = -i$, only the terms corresponding to $n = \text{even} = 2m$ are not cancelled; thus

$$\Theta(1/4|\tau) = \sum_{m=-\infty}^{\infty} q^{4m^2} (-1)^m.$$

On the other hand,

$$\begin{aligned} \Pi(1/4|\tau) &= \prod_{m=1}^{\infty} (1 - q^{2m})(1 + iq^{2m-1})(1 - iq^{2m-1}) \\ &= \prod_{m=1}^{\infty} (1 - q^{2m})(1 + q^{4m-2}) \\ &= \prod_{n=1}^{\infty} (1 - q^{4n})(1 - q^{8n-4}), \end{aligned}$$

where the last line is obtained by considering separately the two cases $2m = 4n - 4$ and $2m = 4n - 2$ in the first factor. Hence

$$(4) \quad c(\tau) = \frac{\sum_{n=-\infty}^{\infty} (-1)^n q^{4n^2}}{\prod_{n=1}^{\infty} (1 - q^{4n})(1 - q^{8n-4})},$$

and combining (3) and (4) establishes our claim that $c(\tau) = c(4\tau)$. Successive applications of this identity give $c(\tau) = c(4^k\tau)$, and since $q^{4^k} = e^{i\pi 4^k\tau} \rightarrow 0$ as $k \rightarrow \infty$, we conclude from (2) that $c(\tau) = 1$. This proves the theorem.

The product formula for the function Θ specializes to its variant $\theta(\tau) = \Theta(0|\tau)$, and this provides a proof that θ is non-vanishing in the upper half-plane.

Corollary 1.4 *If $\text{Im}(\tau) > 0$ and $q = e^{\pi i\tau}$, then*

$$\theta(\tau) = \prod_{n=1}^{\infty} (1 - q^{2n})(1 + q^{2n-1})^2.$$

Thus $\theta(\tau) \neq 0$ for $\tau \in \mathbb{H}$.

The next corollary shows that the properties of the function Θ now yield the construction of an elliptic function (which is in fact closely related to the Weierstrass $\wp$ function).

Corollary 1.5 *For each fixed $\tau \in \mathbb{H}$, the quotient*

$$(\log \Theta(z|\tau))'' = \frac{\Theta(z|\tau)\Theta''(z|\tau) - (\Theta'(z|\tau))^2}{\Theta(z|\tau)^2}$$

is an elliptic function of order 2 with periods 1 and τ , and with a double pole at $z = 1/2 + \tau/2$.

In the above, the primes ' denote differentiation with respect to the z variable.

Proof. Let $F(z) = (\log \Theta(z|\tau))' = \Theta(z|\tau)' / \Theta(z|\tau)$. Differentiating the identities (ii) and (iii) of Proposition 1.1 gives $F(z+1) = F(z)$, $F(z+\tau) = F(z) - 2\pi i$, and differentiating again shows that $F'(z)$ is doubly periodic. Since $\Theta(z|\tau)$ vanishes only at $z = 1/2 + \tau/2$ in the fundamental parallelogram, the function $F(z)$ has only a single pole, and thus $F'(z)$ has only a double pole there.

The precise connection between $(\log \Theta(z|\tau))''$ and $\wp_\tau(z)$ is stated in Exercise 1.

For an analogy between Θ and the Weierstrass σ function, see Exercise 5 of the previous chapter.

1.1 Further transformation laws

We now come to the study of the transformation relations in the τ -variable, that is, to the modular character of Θ .